Lab 01 Template

Answer the first three questions in a pdf file and upload it to Canvas. Questions 4 -6 remind you to upload your python code.

1.) **For Part 01, what's the name of a Shift by N cipher with a key of 3?**
(5 points)

Ans) Caesar's Cipher

2.) **For Part 02, include a screenshot in your lab report.**
(5 points)

```
cpre3310@cpre3310:~/homework/lab01/part02$ python3 part02_skel.py
What character do you want to check?  A
List [0]: 9
List [1]: 0
List [2]: 11
cpre3310@cpre3310:~/homework/lab01/part02$ python3 part02_skel.py
What character do you want to check?  b
List [0]: 0
List [1]: 0
List [2]: 0
cpre3310@cpre3310:~/homework/lab01/part02$ python3 part02_skel.py
What character do you want to check?  B
List [0]: 5
List [1]: 1
List [2]: 1
```

3.) **Document (output copy and paste/screenshot into pdf file) each potential solution you arrived at for Part 03. For each attempt of the key you will include the N used in the shift and the plaintext output from the N. *Did you shift the key LEFT and the resulting characters RIGHT?***
(15 points)

```
cpre3310@cpre3310:~/homework/lab01/part03$ python3 part03_skel.py ciphertext.txt
Testing shift by 0:
PARWBWMAXVHFINMXKLGXXSXUXVTNLXBMATWTOBKNL

Testing shift by 1:
QBSXCXNBYWIGJONYLMHYYTYVYWUOMYCNBUXUPCLOM

Testing shift by 2:
RCTYDYOCZXJHKPOZMNIZZUZWZXVPNZDOCVYVQDMPN

Testing shift by 3:
SDUZEZPDAYKILQPANOJAAVAXAYWQOAEPDWZWRENQO

Testing shift by 4:
TEVAFAQEBZLJMRQBOPKBBWBYBZXRPBFQEXAXSFORP

Testing shift by 5:
UFWBGBRFCAMKNSRCPQLCCXCZCAYSQCGRFYBYTGPSQ

Testing shift by 6:
VGXCHCSGDBNLOTSDQRMDDYDADBZTRDHSGZCZUHQTR

Testing shift by 7:
WHYDIDTHECOMPUTERSNEEZEBECAUSEITHADAVIRUS

Testing shift by 8:
XIZEJEUIFDPNQVUFSTOFFAFCFDBVTFJUIBEBWJSVT

Testing shift by 9:
YJAFKFVJGEQORWVGTUPGGBGDGECWUGKVJCFCXKTWU

Testing shift by 10:
ZKBGLGWKHFRPSXWHUVQHHCHEHFDXVHLWKDGDYLUXV
```

```
Testing shift by 11:
ALCHMHXLIGSQTYXIVWRIIDIFIGEYWIMXLEHEZMVYW

Testing shift by 12:
BMDINIYMJHTRUZYJWXSJJEJGJHFZXJNYMFIFANWZX

Testing shift by 13:
CNEJOJZNKIUSVAZKXYTKKFKHKIGAYKOZNGJGBOXAY

Testing shift by 14:
DOFKPKAOLJVTWBALYZULLGLILJHBZLPAOHKHCPYBZ

Testing shift by 15:
EPGLQLBPMKWUXCBMZAVMMHMJMKICAMQBPILIDQZCA

Testing shift by 16:
FQHMRMCQNLXVYDCNABWNNINKNLJDBNRCQJMJERADB

Testing shift by 17:
GRINSNDROMYWZEDOBCXOOJOLOMKECOSDRKNKFSBEC

Testing shift by 18:
HSJOTOESPNZXAFEPCDYPPKPMPNLFDPTESLOLGTCFD

Testing shift by 19:
ITKPUPFTQOAYBGFQDEZQQLQNQOMGEQUFTMPMHUDGE

Testing shift by 20:
JULQVQGURPBZCHGREFARRMRORPNHFRVGUNQNIVEHF

Testing shift by 21:
KVMRWRHVSQCADIHSFGBSSNSPSQOIGSWHVOROJWFIG
```

```
Testing shift by 22:
LWNSXSIWTRDBEJITGHCTTOTQTRPJHTXIWPSPKXGJH

Testing shift by 23:
MXOTYTJXUSECFKJUHIDUUPURUSQKIUYJXQTQLYHKI

Testing shift by 24:
NYPUZUKYVTFDGLKVIJEVVQVSVTRLJVZKYRURMZILJ

Testing shift by 25:
OZQVAVLZWUGEHMLWJKFWWRWTWUSMKWALZSVSNAJMK
```

4.) Document in the pdf what the plaintext message is. *Did you shift the key LEFT and the resulting characters RIGHT?*
(15 points)

Ans) Plaintext message: WHYDIDTHECOMPUTERSNEEZEBECAUSEITHADAVIRUS

```
Testing shift by 7:
WHYDIDTHECOMPUTERSNEEZEBECAUSEITHADAVIRUS
```

Yes, over here, to decrypt the message, we shifted the letters left by 7; equivalently, since 26-7 = 19, we can also say we shifted the characters right by 19.

5.) **Submit your commented code from Part one as lab01_part01.py to canvas**
(20 points)

6.) **Submit your commented code from Part two as lab01_part02.py to canvas**
(20 points)

7.) **Submit your commented code from Part three as lab01_part03.py to canvas** ***<u>Did you shift the key LEFT and the resulting characters RIGHT?</u>***
(20 points)

Ans) Over here, we are basically using a brute force approach to decrypt the ciphertext by trying all the possible Caesar cipher shifts, and it basically subtracts the shift amount from each character's ASCII value (wrapping around A-Z), for each shift's value, which reverses the original encryption effectively.

So, over here I basically inspected each output and figured out that shift 7 produced a meaningful English sentence, which was, "WHY DID THE COKMPUTER SNEEZE BECAUSE IT HAD A VIRUS", which basically confirmed that the original message had been encrypted with a Caesar cipher using a shift value of 7.

Over here, I also learned that Caesar ciphers can be broken using brute force, since there are only 26 possibilities. I also learned how to manipulate each character in Python.